

Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – 2nd part (Parallel Programming)–September 7th 2022

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Questions 1 (4 points) _____

Questions 2 (4 points) _____

Questions 3 (4 points) _____

Questions 4 (4 points) _____

Questions n. 1

Answer the following questions about OpenMP and briefly explain (without explanation the answer will be considered invalid)

A. What are the static and dynamic extents of the omp parallel directive?

B. Is it possible for the programmer to control the distribution of parallel loop iterations across threads when using an omp for directive?

C. What is the functionality of the omp_set_num_threads function? Are there other ways for achieving the same result?

D. Is it possible to perform a reduction over a single variable using OpenMP?

Questions n. 2

Answer the following questions about parallel algorithms and parallel programming and briefly explain (without explanation the answer will be considered invalid)

A. Is it possible to use multiple parallelization models in the same program? Would it benefit some hardware architectures in particular?

B. Describe the scan parallel pattern. Can it be implemented using different approaches with respect to the computation of intermediate values?

C. Which are the main differences between parallel programming models based on threads and those based on processes in terms of low-level information management and architectures that can benefit from them?

Questions n. 3

Answer the following questions about MPI and briefly explain (without explanation the answer will be considered invalid)

A. Describe the processes synchronization model used by MPI.

B. Describe the behavior of the MPI_Bcast routine.

C. Describe the use of communicators in MPI and the advantages of defining multiple communicators in a program.

Questions n. 4

Answer the following questions about PThread and briefly explain (without explanation the answer will be considered invalid)

A. What is the use of condition variables in PThreads?

B. Is it possible to share work over a fixed size thread pool that is spawned once at the beginning of the program execution? What kind of parallel pattern can be implemented in this way?